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Document: Electrical ICD MS30-0102 Payload & Hive2		
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Interface Control Document (ICD) **For Micro 300-102 Payload & Hive2**

Payload PN: MS30-0102-00
Hive2 PN: HVE2-0101-00 / HVE2-0102-00



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Changes and Revisions

REV	EO/ECO	Description	Date
-	EC25001156	Initial Distribution for MS30 0101 ICD	21 July 2025
A	EC26000436	Update to payload's power connector P/N and pinout. Updated HIVE picture. Corrected speed units for MAX power consumption.	31 March 2026

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1.1 Payload power supply

	MIN	Nominal	Max ISR	Max LM @120Knots	UNIT
Input voltage	24	28	32	32	V[dc]
Power consumption	10	20	50	100	W
Inrush current	700				mA/2mS

1.2 Hive2 power supply

	MIN	Nominal	MAX	UNIT
Input voltage	12	28	36	V[dc]
Power consumption	10	12	15	W
Inrush current			1	A/50uS

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2. Payload interface

2.1 Payload power connector description

The payload power connector is via J2 connector - 2 positions

Main Connector	System	Mating
P/N	M80-5000442	M80-4140498
RefDes	J2	P2

The payload power connector is used for input power

2.1.1 Payload power connector pin-out

Pin #	Signal	Destination
1	28v RTN	Payload Power Supply RTN
2	+28v	Payload Power Supply
3	28v RTN	Payload Power Supply RTN
4	+28v	Payload Power Supply

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2.2 Payload USB-C connector description

The payload interface is via J1 connector - USB Type C receptacle

Main Connector	System	Mating
P/N	USB C receptacle	USB C Plug
RefDes	J1	P1

The Payload USB-C connector is used for data input & output

2.2.1 Payload USB-C connector pin-out

A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1
GND	RX2+	RX2-	V BUS	SBU1	D-	D+	CC	V BUS	TX1-	TX1+	GND
GND	TX2+	TX2-	V BUS	V CONN			SBU2	V BUS	RX1-	RX1+	GND
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12

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3. HIVE2 interface

3.2 Hive2 Power connector description

Hive2 power connector is via JST connector - 6 positions

Main Connector	System	Mating
P/N	SM06B-GHS-TBT	GHR-06V-S
RefDes	PWR	

The Hive2 power connector is used for input power

3.2.1 Hive2 Power connector pin-out

Pin #	Signal	Destination
1	+28v	External Power Supply
2	+28v	External Power Supply
3	+28v	External Power Supply
4	28V RTN	External Power Supply RTN
5	28V RTN	External Power Supply RTN
6	28V RTN	External Power Supply RTN

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3.3 Hive2 USB-C connector description

Hive2 communicates with the payload through a Type C cable.

Type C Connector	System	Mating
P/N	Type C receptacle (with dual screw locking option)	Type C plug
RefDes	USB-C	

3.3.1 Hive2 USB-C connector pin-out

A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1
GND	RX2+	RX2-	V BUS	SBU1	D-	D+	CC	V BUS	TX1-	TX1+	GND
GND	TX2+	TX2-	V BUS	V CONN			SBU2	V BUS	RX1-	RX1+	GND
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12

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3.4 Hive2 Ethernet connector description

Hive2's main interface is via JST connector

GPIO Connector	System	Mating
P/N	SM08B-GHS-TB	GHR-08V-S
RefDes	ETH	

The connector is used for ethernet link.

3.4.1 Hive2 Ethernet connector pin-out

Pin #	Signal	Destination
1	ETHERNET MDIO_P	Primary Output: host - RJ45_pin2
2	ETHERNET MDIO_N	Primary Output: host - RJ45_pin1
3	ETHERNET MDI1_P	Primary Output: host - RJ45_pin6
4	ETHERNET MDI1_N	Primary Output: host - RJ45_pin3
5	ETHERNET MDI2_P	Primary Output: host - RJ45_pin5
6	ETHERNET MDI2_N	Primary Output: host - RJ45_pin4
7	ETHERNET MDI3_P	Primary Output: host - RJ45_pin8
8	ETHERNET MDI3_N	Primary Output: host - RJ45_pin7

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3.5 Hive2 Serial connectors description

Hive2 serial connectors is via 3 JST connectors – 4/6 positions each.

The serial connectors are used for communications, discrete I/O and maintenance signals.

Serial0:

Main Connector	System	Mating
P/N	SM04B-GHS-TBT	GHR-04V-S
RefDes	SER0	

3.5.1 Hive2 SER0 connector pin-out

Pin #	Signal	Destination
1	+5v	+5v
2	UART1_TXD	(logic 3.3v) / RS232_TX
3	UART1_RXD	(logic 3.3v) / RS232_RX
4	GND	GND

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Serial1:

Main Connector	System	Mating
P/N	SM06B-GHS-TBT	GHR-06V-S
RefDes	SER1	

3.5.2 Hive2 SER1 pin-out

Pin #	Signal	Destination
1	+5v	+5v
2	UART1_TXD+	(logic 3.3v) / RS422_TX+
3	UART1_TXD-	(logic 3.3v) / RS422_TX-
4	UART1_RXD+	(logic 3.3v) / RS422_RX+
5	UART1_RXD-	(logic 3.3v) / RS422_RX-
6	GND	GND

Serial2:

Main Connector	System	Mating
P/N	SM04B-GHS-TBT	GHR-04V-S
RefDes	SER2	

3.5.3 Hive2 SER2 pin-out

Pin #	Signal	Destination
1	+5v	+5v
2	UART2_RXD	logic 3.3v
3	UART2_TXD	logic 3.3v
4	GND	GND

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3.6 Hive2 USB1 connector description

Hive2 USB1 connector is via USB1 connector - 4 positions

Main Connector	System	Mating
P/N	SM04B-GHS-TBT	GHR-04V-S
RefDes	USB1	

3.6.1 Hive2 USB1 connector pin-out

Pin #	Signal	Destination
1	5V	5V
2	USB1_D-	USB1_D-
3	USB1_D+	USB1_D+
4	GND	GND

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3.7 Hive2 USB2 connector description

Hive2 USB2(OTG) connector is via USB2connector - 8 positions

Main Connector	System	Mating
P/N	SM08B-GHS-TBT	GHR-08V-S
RefDes	USB2	

3.7.1 Hive2 USB2 connector pin-out

Pin #	Signal	Destination
1	5V	OTG_5V (input)
2	USB2_D-	USB2_D-
3	USB2_D+	USB2_D+
4	USB2_SSTX-	USB2_SSTX-
5	USB2_SSTX+	USB2_SSTX+
6	USB2_SSRX-	USB2_SSRX-
7	USB2_SSRX+	USB2_SSRX+
8	GND	GND

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4. Video interface and formats

4.1 Main Video Output – Ethernet.

Hardware layer: 10/100 Ethernet Hive2 ETH connector.

Video Layer: Video output compressed in H.265 format.

Frames per second: 30.

DTV Resolution: 1920x1080.

TI video frame: 1920x1080 (sensor resolution 640x480).

5. Connectors datasheet



SM0xB-Datasheet.pdf